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The hierarchical morphotope classification: A theory-driven framework for large-scale analysis of built form

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ABSTRACT

Built environment, formed of a plethora of patterns of building, streets, and plots, has a profound impact on how cities are perceived and function. While various methods exist to classify urban patterns, they often lack a strong theoretical foundation, are not scalable beyond a local level, or sacrifice detail for broader application. This paper introduces the Hierarchical Morphotope Classification (HiMoC), a novel, theory-driven, and computationally scalable method of classification of built form. HiMoC operationalises the idea of a morphotope – the smallest locality with a distinctive character – using a bespoke regionalisation method SA^3 (Spatial Agglomerative Adaptive Aggregation), to delineate contiguous, morphologically distinct localities. These are further organised into a hierarchical taxonomic tree reflecting their dissimilarity based on morphometric profile derived from buildings and streets retrieved from open data, allowing flexible, interpretable classification of built fabric, that can be applied beyond a scale of a single country. The method is tested on a subset of countries of Central Europe, grouping over 90 million building footprints into over 500,000 morphotopes. The method extends the capabilities of available morphometric analyses, while offering a complementary perspective to existing large scale data products, which are focusing primarily on land use or use conceptual definition of urban fabric types. This theory-grounded, reproducible, unsupervised and scalable method facilitates a nuanced understanding of urban structure, with broad applications in urban planning, environmental analysis, and socio-spatial studies.

1. Introduction

The structure of the built environment affects how cities, towns and villages are perceived as well as the types of activities they can host. The way buildings are laid out, the patterns of streets, the unique open spaces and transformations we observe when we move between neighbourhoods create experiences allowing us to identify a place (Lynch, 1960), navigate through urban spaces (Bongiorno et al., 2021) and influence housing preferences and choices (Hasanzadeh et al., 2019). Urban morphology, a field that primarily focuses on understanding of built up patterns and their evolution (Moudon, 1997), has a long history in unpacking the complexity of urban (or general built up, when dealing with rural and peri-urban locations) form. While approaches vary from architectural interpretation (Caniggia & Maffei, 2001) to modelling based on remote sensing (Abascal et al., 2022), the overall goal is to identify patterns of development in the built fabric. Attempts to do this range from the detection of local morphological regions (Conzen, 1988), prediction of planning archetypes (Chen et al., 2021), or delineation of unique concepts like fringe belts (Whitehand, 1967) or informal settlements (Wang et al., 2023). Recently, there has been increased interest in one specific approach – exhaustive classification frameworks.

Researchers use classification methods to reduce the complexity of the built-up environment into types. While they can be both supervised (classes are known prior the analysis) or unsupervised (classes are a result of the analysis), the resulting typologies are used to better understand the structure of cities and their relationship to other phenomena. For example, the built-up environment's impact on the heat island effect (Wu et al., 2025), energy efficiency (Zhu, Li, et al., 2022), broader economic outcomes (Wang & He, 2024), as well as the overall patterns of urbanisation (Domingo et al., 2023; Wang & He, 2024). Furthermore, analysing the interplay of the physical urban fabric with land use (Fleischmann & Arribas-Bel, 2022), mobility (Calafiore et al., 2023), vegetation (Guyot et al., 2021) or other dimensions of activities is more manageable when built form can be represented by a small number of

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interpretable types. However, each of these application domains typically has a different classification method focused on different aspects and scales. Some focus only on characteristics and configurations of buildings (Caruso et al., 2017); or only on streets (Araldi & Fusco, 2024); or on total built-up surface within a predefined unit, i.e. patch (Jochem et al., 2020).

These classifications of urban environment have a long history across disciplines and can involve both qualitative and quantitative aspects. Urban planning approaches with a pronounced qualitative component typically provide a detailed account of a small geographical area, based on the particular characteristics of the urban fabric in the area, including social aspects, as well as its history of development (Conzen, 1960). Driven by advancements in data processing and computing, recent works have successfully adopted quantitative methods to achieve similar hyper-locality and detail (Araldi & Fusco, 2019, 2024; Fleischmann et al., 2022). Such approaches create detailed typologies that differentiate between different historical layers, distinguishing medieval from modernist urban developments, single family housing from row-houses and similar (Arribas-Bel & Fleischmann, 2022; Fleischmann et al., 2022).

Nevertheless, a consistent morphological classification that stretches beyond a delimited boundary of a single metropolitan area or a country is still missing. At least three factors have prevented such work being carried out so far – theory, data and methods – each interacting with the others and amplifying the overall difficulty of the task. Theoretical considerations drive applications and both data and methodological choices. However, they are at the same time limited by data availability and methodological scalability. Therefore, theoretically relevant approaches are typically limited by their method to neighbourhoods or multiple cities at most, i.e. (Dibble et al., 2019; Fleischmann et al., 2022). Other purely quantitative approaches generally sacrifice detailed typologies and theoretical justifications, and focus only on a few coarse morphological features in favour of scalability to larger regions (European Environment Agency, 2020, 1990; Stewart & Oke, 2012).

This paper aims to tackle the three interrelated issues of theory, data and methods by creating a scalable urban morphology classification method that is directly stemming from theoretical foundations of urban planning.

The rest of the paper is structured as follows. The next section positions the overall quantitative urban morphological approach, as well as our specific method, within the literature. The Methods and Data sections describe the approach we have taken in detail, and also provide some background information about the study area. Furthermore, they describe the qualitative and quantitative steps we take to verify the resulting classification – the descriptions of types, the spatial distribution of the results and the quantitative comparisons with other available large scale urban form datasets. We present the outcomes in the Results section, while the Discussion and Conclusions section focuses on the implications of the results, their limitations and the potential for future work.

2. Conceptualisation of urban form

Within cities conceptualised as complex adaptive systems (Batty, 2012), urban form is a component that typically reflects the configuration of buildings, streets and plots within the built-up landscape (Moudon, 1997). These, so called, fundamental elements of urban form have been present in most of the early schools of urban morphology, no matter if they were stemming from architecture (Caniggia & Maffei, 1979; Muratori, 1959) or geography (Conzen, 1960; Whitehand, 1967). Researchers have been attempting to group and delineate them to allow discussion about a *type* of urban form and its spatial allocation, since theoretically, there are infinitely many possible configurations. Conzen's *morphological region* (Conzen, 1975) or Italian *urban tissue* (Muratori, 1959) are among the concepts defining a morphologically homogeneous patch of land, that can be further compared to other such patches, eventually identifying a recurring type. Each uses its own criteria and methods of delineation, but share the same goal. Importantly, Conzen further suggests there is a hierarchy of morphological regions (in a sense of nesting rather than ranking) with the smallest one being called *morphotope* (as in *biotope*), defined as "*[t]he smallest urban localit[y] obtaining distinctive character among their neighbours […]*" (Conzen, 1988, 259). This is typically closely linked to the period-determined building types and requires information on the process of historical development within the study area (Arat, 2023; Conzen, 2018). An analysis with this type of data dependency is therefore cumbersome to scale beyond individual cities as the data capturing development period in which a building has been constructed is scarce and often encoded only in digitised scans of historical maps. At the same time, given the definition of "the smallest" is not easy to define, researchers tend to skip the detection of morphotopes and focus primarily on morphological regions of higher order with a handful of works (e.g. Fedchenko (2023)) aiming to identify morphotopes within urban fabric. Yet, the idea is fundamental and the identification of morphotopes and their hierarchy of morphological regions at different levels of granularity could be used within a wide range of applications, both within and outside urban morphology.

While Conzen's methods are primarily qualitative, based on the interpretation of historical maps and town plans (Oliveira, 2019), quantitative approaches are on the rise trying to resolve some of the issues with scalability and reproducibility of morphological analysis. Among such methods, morphometric analysis, stemming from early works of Conzen (1988) or Slater (1981) is gaining more prominence.

Morphometric analysis, which has been originally suggested for numerical analysis of plots based on their detailed measurements (e.g. width in Slater (1981)), has grown into a distinct area of study called *urban morphometrics* (Oliveira & Porta, 2025). While the term has been around at least since Porta et al. (2011), it lacks a formal established definition. Here, we offer one and define urban morphometrics as "*a study of urban form through the means of quantitative assessment of its constituent elements*", where constituent elements are typically derived from Moudon's fundamental ones but are further influenced by data availability and the aims of the study. At the same time, the links between morphological theory and quantitative computational methods proposed under the umbrella of urban morphometrics (explicitly stated or not) do not tend to be particularly strong. There are a few exceptions, notably the work of Araldi and Fusco (2019, 2024) stemming from the combination of British and French theories of urban morphology mixed with the streetscape skeleton approach or that of Fleischmann et al. (2022) building on a mixture of British and Italian foundations. We believe that a morphometric assessment needs to be well founded within morphological theory and that such a link should be made explicit, rather than employing the *"data will tell"* approach hoping that the measurement framework is accidentally sensible. Only in this way can we minimise the bias of the morphometric method and make it trustworthy and interpretable.

The methods of urban classification that can be considered morphometric are a relatively recent development, with the earliest attempts dated to late 2000s and early 2010s (Colaninno et al., 2011; Gil & Montenegro, 2012; Song & Knaap, 2007). While some other related works could be listed here, they usually do not fulfil the definition of urban morphometrics, typically due to reliance on different features than constituent elements of urban form, typically pixels or patches (Huang et al., 2007; Thomas et al., 2010; Wurm et al., 2010), or are very borderline in their methods (Smith & Crooks, 2010). Since then, the number of methods have been consistently growing, showcasing the attractiveness of urban morphometrics that is partially driven by the rise in the availability of data capturing morphological elements (e.g. growth of OpenStreetMap (Biljecki et al., 2023)) and partially by the growth of geographic data science (Singleton & Arribas-Bel, 2019).

Morphometric classifications started on sub-city scales, focusing on individual neighbourhoods (Gil & Montenegro, 2012) or transects

through the city (Colaninno et al., 2011). Later, they proceeded to cover entire cities or metropolitan regions (Araldi & Fusco, 2019; Bobkova et al., 2019; Fleischmann et al., 2022; Jochem et al., 2020; Schirmer & Axhausen, 2015). Only recently, the evolution of methods, availability of computational resources and data resulted in multiple methods that cover entire countries (Araldi & Fusco, 2024; Fleischmann & Arribas-Bel, 2022; Jochem & Tatem, 2021). Beyond that scale, classifications of urban form do exist (Demuzere et al., 2022; European Environment Agency, 2020) but are typically not of a morphometric origin and a result of remote sensing or other procedures. That pushes them away from morphological theory, yielding classifications that are often too simplistic or archetypal due to a requirement of classes to be defined prior to running the analysis, unlike in a typical morphological assessment where classes stem from observations on the ground.

Methodologically, we can observe rising complexity of morphometric classification methods. First attempts were using a handful of measurements, typically followed by k-Means clustering or similar (Bobkova et al., 2019; Gil & Montenegro, 2012), occasionally replaced by more robust clustering algorithms and a higher number of measurements (Araldi & Fusco, 2019; Fleischmann et al., 2022). While these approaches remain popular due to their simplicity and good overall performance (Marwal & Silva, 2023; Venerandi et al., 2024; Yu et al., 2023), methods derived from neural networks appeared (Chen et al., 2025; De Sabbata et al., 2023; Liu & Song, 2025) in an attempt to extract patterns in a more advanced way. Despite these developments, large scale classifications are still hard to come by and those that exist cannot be classified as morphometric.

Typically, global or continental data products make use of Local Climate Zones (LCZ) (Stewart & Oke, 2012), which in the case of urban form contain 10 conceptually defined classes. Thanks to the progress in remote sensing, these classes can be predicted across the globe (Demuzere et al., 2022; Zhu, Qiu, et al., 2022) and then further analysed (Debray et al., 2025) but LCZs are a rough proxy of intricacies of urban form at best, reiterating the point above about the missing link between morphological theory and classification offered. Another approach to large scale morphological studies is shown by Biljecki and Chow (2022), who provide a limited set of morphometric indicators without an attempt at classification, though on a global scale, or in Li et al. (2021) providing a similar dataset aggregated to 100 m grid for selection of Chinese cities. Furthermore, Moosavi (2022) provides a global morphological analysis at a neighborhood level, rather than attempting more granular ones.

The current state of morphological and morphometric classifications can henceforth be summarised as follows. The methods that are strongly embedded in theory tend to be not scalable and typically qualitative. The methods that are quantitative (morphometric) and theory-driven (at least partially) have seen an improvements in their ability to scale to national extents but none have been tested on a larger area, raising questions on where their limits are. The methods that show scalability to global extents lack the theoretical foundation in urban morphology and a detail we require from morphological classifications.

The aim of this paper is to develop a method of morphometric classification that is heavily theory-driven, building on Conzen's concepts of morphotope and hierarchical embedding of morphological regions, but completely computational and scalable beyond national boundaries at the same time. To achieve this, we propose a *"Hierarchical Morphotope Classification"* (HiMoC) – a hierarchical method for classifying urban form into a taxonomic tree of algorithmically delineated morphotopes.

3. Methods

The HiMoC method can be broken down in eight steps as illustrated in Fig. 1. Starting with data retrieval and pre-processing, followed by morphometric characterisation, delineation of morphotopes and their morphometric characterisation, closing with definition of morphological linkage and flexible understanding of urban fabric. Each of the steps is described in detail in the following sections.

3.1. Urban form as data

Any morphometric assessment first needs to represent the constituent elements of urban form as data to allow measurement. In our case, we limit the input to two datasets (Fig. 1 a). First, building footprints represented as polygons, and second, street networks represented as lines capturing individual road segments. This input captures only a subset of morphological properties (e.g., there is no information on plot boundaries or open space) but is relatively abundant, ensuring that the data availability is not the main limiting factor in applying the method to different contexts. At the same time, this input proves to capture the core essence of morphological patterns (Araldi & Fusco, 2019; Fleischmann et al., 2022). The omission of building height data might hinder certain aspects of the classification. However, we argue that the absence of height data is not critical since 2-dimensional morphology tends to capture height information (Milojevic-Dupont et al., 2020).

The foundational spatial element on which the classification is done is a building, as the smallest available morphological feature we can work with. From streets and buildings, we further generate enclosed tessellation (Arribas-Bel & Fleischmann, 2022), which partitions space into regions delimited by street geometry and distance to buildings. Formally, it is *"the portion of space that results from growing a morphological tessellation within an enclosure delineated by a series of natural or*

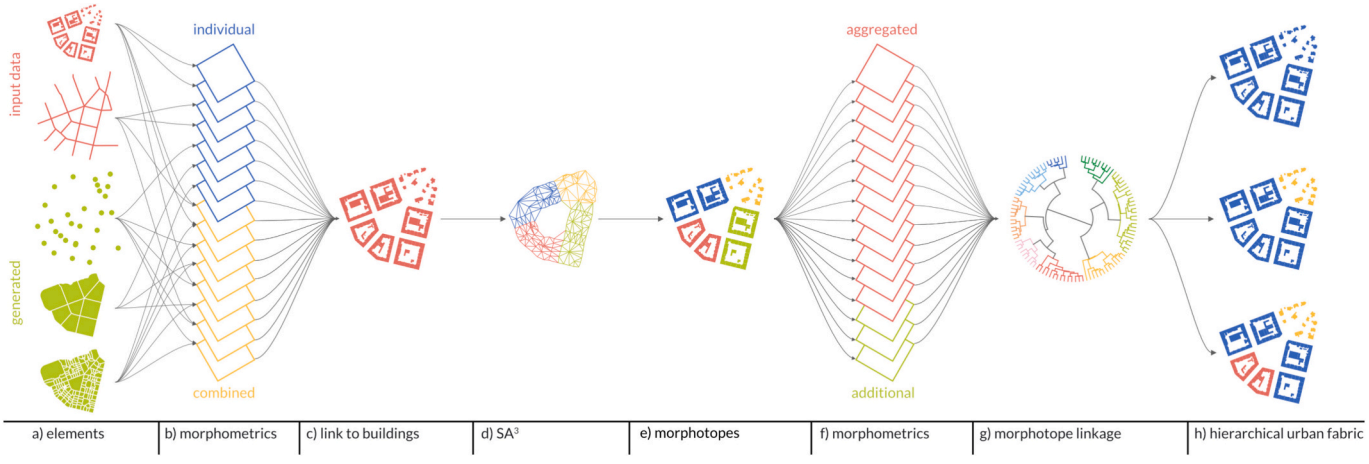


Fig. 1. Illustration of the methodology.

built barriers identified from the literature on urban form, function and perception" (Arribas-Bel & Fleischmann, 2022, 5), where morphological tessellation is a polygon-based Voronoi tessellation with building footprints as an input (Fleischmann et al., 2020). Enclosed tessellation serves as a proxy for the calculation of morphometric attributes on a plot level while, importantly, capturing the open spaces between buildings.

Compared to previous applications of enclosed tessellation, we are not interested in unbuilt open spaces. Therefore we limit the extent of the enclosed tessellation around each building using a variable bandwidth. The bandwidth reflects the assumed size of an underlying plot based on distance to surrounding buildings and is different for each enclosed tessellation cell (see *Appendix 15*).

The street network is split into nodes, representing junctions, and edges, representing segments, as different morphometric attributes are computed for each. The data is further processed to ensure topological correctness of the building layer and simplification of the street network to reflect a morphological, rather than a transportation representation, based on Fleischmann et al. (2025). For the details, refer to *Appendix 13*.

3.2. Morphometric characterisation

Measurements are at the core of every morphometric method. While there are potentially hundreds of attributes to measure, not all are equally relevant. Many capture the same concept (e.g. various shape indices reported in Basaraner and Cetinkaya (2017)) or are useful for analysis at scales we are not interested in (e.g. Schirmer and Axhausen's (2015) accessibility metrics). The guidelines proposed by Sneath and Sokal (1973) indicate that the selection should be as inclusive as possible while avoiding conceptual duplications (among other). A selection of this sort has already been proposed in Fleischmann et al. (2022) and given we share the same input data and general aim, our selection starts from their set. We only remove those depending on height as that is not available at scale and add a small subset of additional attributes capturing several concepts the original set did not.

In total, we measure 59 attributes reported in detail in *Appendix 10* and implemented in a Python package momepy (Fleischmann, 2019) (Fig. 1 b). The set is designed to balance properties of streets, buildings, street intersections and enclosed tessellation cells themselves at three scales: *small*, covering only aspects of the element; *medium*, covering aspects of the element and neighbouring elements; and *large*, covering neighbouring elements up to five topological neighbours; and two subsets: the *individual* layer (e.g. measuring a shared walls ratio of buildings) and the *combination* of layers (e.g. measuring the relation of a building to a street). Each character is measured at the level of individual elements (be it building, street node, or tessellation cell) and subsequently linked to the tessellation layer (Fig. 1 c). At every step of the process, all attributes have an equal weight.

3.3. Morphotope delineation

Delineating morphotopes based on the morphological attributes of buildings and their surroundings as outlined in Section 3.2 is the first type of aggregation we carry out. Going back to the definition of a morphotope, our goal is to find the smallest urban locality obtaining distinctive character. That implies a strict requirement of contiguity of a morphotope and an application of a clustering method able to define boundaries. To operationalise this step we propose a bespoke regionalisation method called Spatial Agglomerative Adaptive Aggregation (SA^3). The algorithm generates a full linkage matrix using Ward's agglomerative clustering restricted to make connections on a contiguity graph of tessellation cells linked to buildings. Then, we extract clusters from the linkage using the density-based cluster extraction method called 'leaf' (McInnes et al., 2017) (Fig. 1 d, e). The specific implementation details are provided in *Appendix 11*.

The choice of this algorithm for the delineation of morphotopes is guided by theoretical and computational considerations. First, it produces spatially contiguous clusters. Second, the algorithm admits noise – it can assign a building to a noise cluster if there is not enough of its immediate neighbours in geographical space that are morphologically similar. Third, there can be more morphological heterogeneity within some morphotopes, than between neighbouring morphotopes of different types. For example, a medieval or industrial area has more internal heterogeneity, than two neighbouring morphotopes that represent semi-detached and single family houses respectively. Due to the usage of density-based extraction, SA^3 can delineate these types of morphotopes despite their variable internal heterogeneity. In contrast, methods that rely on a flat cut of the full linkage matrix could either split some types of morphotopes if the cut value is too low, or combine them if the value is too high. Fourth, the final number of morphotopes does not need to be specified a priori. The only parameter required by SA^3 is the minimum number of buildings to form a morphotope. The optimal value was determined empirically using Total Within-Cluster Sum of Squares analysis reported in *Section 19*. Based on that, we selected a value of 75 as reflecting the definition of the morphotopes the closest. The implementation of SA^3 has been contributed to a Python package spopt (Feng et al., 2022) and used from there.

Taken all together, our operationalisation of the concept of a morphotope is: a group of 75 or more directly neighbouring buildings that are more similar to each other morphologically, than to any other such group around them.

3.4. Morphotope linkage

Once the morphotopes are delineated, the remaining task is to capture the dissimilarities between them in order to generate a nested hierarchy. This can be done by using the original morphometric attributes that went into SA^3 aggregated per morphotope. We use median value of each, while measuring three additional attributes enabled by the new morphotope aggregation, all outlined in *Appendix 10* (Fig. 1 f). We then combine the morphotopes based on similarity only in feature space (no spatial restrictions) to produce a linkage in the form of a taxonomic tree (Fig. 1 g). This is done using Ward's agglomerative clustering algorithm. To ensure scalability of the algorithm we generate a 10 nearest neighbours graph (in feature space) for the full dataset and limit the potential Ward distance calculations to this subset. This step only marginally affects the top of the tree while significantly reducing the computational requirements.

The linkage only includes morphotopes and all the buildings labelled as noise in the previous delineation step were excluded. In order to include them in the final analysis, we later assign them to a branch of the taxonomy based on morphometric similarity. To do this we first, group together noise buildings into "noise pseudo-morphotopes" based on spatial contiguity. Then we select a flat cut of the hierarchy, where each branch represents a type with attributes equal to the median value of the morphotope features within it. Lastly, each "noise pseudo-morphotopes" is assigned the most common branch, among its five nearest clusters based on similarity in feature space. Alternatively, all noise points can be discarded.

The final hierarchy shows the dissimilarity between morphotopes, which can be extended all the way down to dissimilarity between pairs of buildings since the SA^3 algorithm is also hierarchical. This hierarchy can be cut at arbitrary levels to produce clusters at various scales which represent taxa or urban fabric types, similarly to Conzen's hierarchy of morphological regions. This approach follows the tradition of phenetic trees (those based on physical appearance) (Sneath & Sokal, 1973), and does not attempt to explain any phylogenic relations, despite similar structure.

3.5. Interpretation

The hierarchical tree is a powerful representation of morphological

similarity, but one that is non-trivial to interpret. With flat clusters as in Fleischmann and Arribas-Bel (2022), one needs to only name and describe each. Within a tree, such an exercise needs to be done with the hierarchy in mind and can quickly get out of hand if we go too deep. To facilitate the interpretation of HiMoC, we propose naming branches up to three levels of bifurcation. For each level, we describe each taxa branch based on the morphological features of the morphotopes within them and provide a short description and pen portrait of each. This naming minimises the complexity of the subsequent analysis and highlights the general patterns in the taxonomic structure.

3.6. Evaluation

Evaluating the validity of a classification of this sort is always complicated as there is typically no "ground truth" to compare to. To mitigate this problem to an extent, we use external data capturing adjacent concepts and determine whether the degree of similarity between our results and the external data is sensible or not. We compare the results of HiMoC at our most granular named level with three existing data products – CORINE Land Cover (European Environment Agency, 1990), Copernicus Urban Atlas (European Environment Agency, 2020), and Global Map of Local Climate Zones by Demuzere et al. (2022). Each offers a different perspective on our results. CORINE, with its limited granularity of classes that cover the urban environment, provides insight into relation of taxa to urban continuity and specialised land use (industrial zones). Urban Atlas introduces an explicit dimension of density alongside continuity. LCZs theoretically provide the largest granularity of urban classes (10) that take into account compactness, height, density, size and land use but are, in principle, conceptually defined. The underlying assumption is that there should be a clear relationship between the taxonomy and each of these classifications, limited by the different concepts they are built on. To quantitatively analyse the similarity, we create confusion matrices based on spatial intersections, by assigning each building from our hierarchy the appropriate label from each of the three comparison datasets. Our main goal with the comparisons is to demonstrate the usefulness of our results by showing where our clusters overlap and potentially extend existing data products.

Another level of evaluation can be done qualitatively, using visual observation and expert knowledge. This involves a selection of case studies to focus on and understanding whether the patterns captured at various levels represent expert understanding of morphology of the area. Ideally, the selection shall cover different sizes of settlements and multiple geographic regions within the study area. Here, we will present a subset of this exercise focusing on four settlements from population of 8 thousand to 3.4 million. We analyse the patterns of urban form as we move through the three named levels of the taxonomy and the relative importance of different features.

Furthermore, we analyse the spatial distribution of the most granular named level, in order to demonstrate the ability to capture urban form patterns across administrative borders.

It should be noted that we use flat cuts of all subsequent analysis, but multilevel cuts are possible and likely to produce better results for specific applications.

4. Data

Our study area covers six contiguous European countries – Austria, Czechia, Germany, Lithuania, Poland, and Slovakia. There are three main reasons for this selection – data availability, morphological diversity, and available expert knowledge.

First, there is official open cadastral data available for every one of these countries as they are all part of the INSPIRE program (Craglia & Annoni, 2007) and provide polygons for the entire national building stock in a (relatively) consistent manner. Second, the urban form of Central Europe is highly heterogeneous due to different planning paradigms and approaches applied in history, and due to the presence of well-preserved historical districts as well as those coming from post-war rebuilding. Although not exhaustive, this diversity enables us to evaluate the method on a wide range of urban fabric types. Last, the research team has a large experience of working on morphological research in the region and is able to expertly evaluate the outcomes of the method.

There are two morphological elements required to build the taxonomy – building footprints in the form of two-dimensional polygons and streets in the form of linestrings. The buildings are downloaded from the official cadastres of each country, while the streets are a direct download from Overture Maps. In total, the classification covers over 90 million building footprints and around 21 million street network segments.

5. Results

As indicated in the methods, the processing pipeline generates large amounts of data divided into multiple steps. Here we report results from the key parts – morphotope delineation, linkage, interpretation, and evaluation. For detailed insights into morphometric characterisation, please refer to the data product outlined in Section 7.

5.1. Morphotope delineation

The SA^3 algorithm resulted in the delineation of more than 500,000 morphotopes as spatial aggregation of 90 million buildings. A mean size of a morphotope is 132 buildings, with the largest one covering 636 buildings. An illustrative example covering the city centre of Prague, CZ, is shown in Fig. 2.

The result tends to capture the concept of morphotope very well from the interpretative perspective - the delineations are typically *the smallest locality obtaining distinctive character* and it is unlikely that any further splitting would be meaningful. What Fig. 2 illustrates is the high sensitivity of the algorithm to differences in morphological structure and an excellent ability to follow the edges of distinct patterns, which is something that similar methods typically struggle with (e.g. Fleischmann et al. (2022), Fleischmann and Arribas-Bel (2022), or Araldi and Fusco (2024)). One can argue that some neighbouring morphotopes could be joined. However, having more granular delineations lies closer to the idea of *the smallest* and the subsequent construction of the taxonomic tree will create these further joins. The relative importance of individual morphometric attributes for the delineation is further reported in *Section 18*.

5.2. Morphotope linkage and taxonomy

The linkage between the morphotopes is done using Ward's agglomerative clustering, meaning that the resulting structure is a hierarchical tree capturing dissimilarity between pairs and later groups of morphotopes within its branches. While the full tree has around 500,000 levels, it is typically more practical to look only at the coarsest branches and rely on the most granular ones for specific localised use cases.

Fig. 3 provides a high-level overview of the resulting taxonomy organised around a radial representation of the tree, showing the 26 coarsest branches. The tree is coloured according to the interpretation of the branches at the third bifurcation (see Section 5.3), showing named types of urban fabric. The five rings around the tree provide an insight into a numerical profile of each branch based on a selection of morphometric attributes at the morphotope level. This captures the reasons why the tree looks as it looks and why we consider some morphotopes more similar than others. The legend of attributes further shows the global distribution of values. The next ring represents a bar chart of abundance of each type within the study, represented as the log of the number of morphotopes per branch, highlighting that some types of urban fabric are more prevalent than others and that certain branches can even be considered as extremely rare outliers. The final two rings contain two morphotopes nearest to the centroid of each branch and

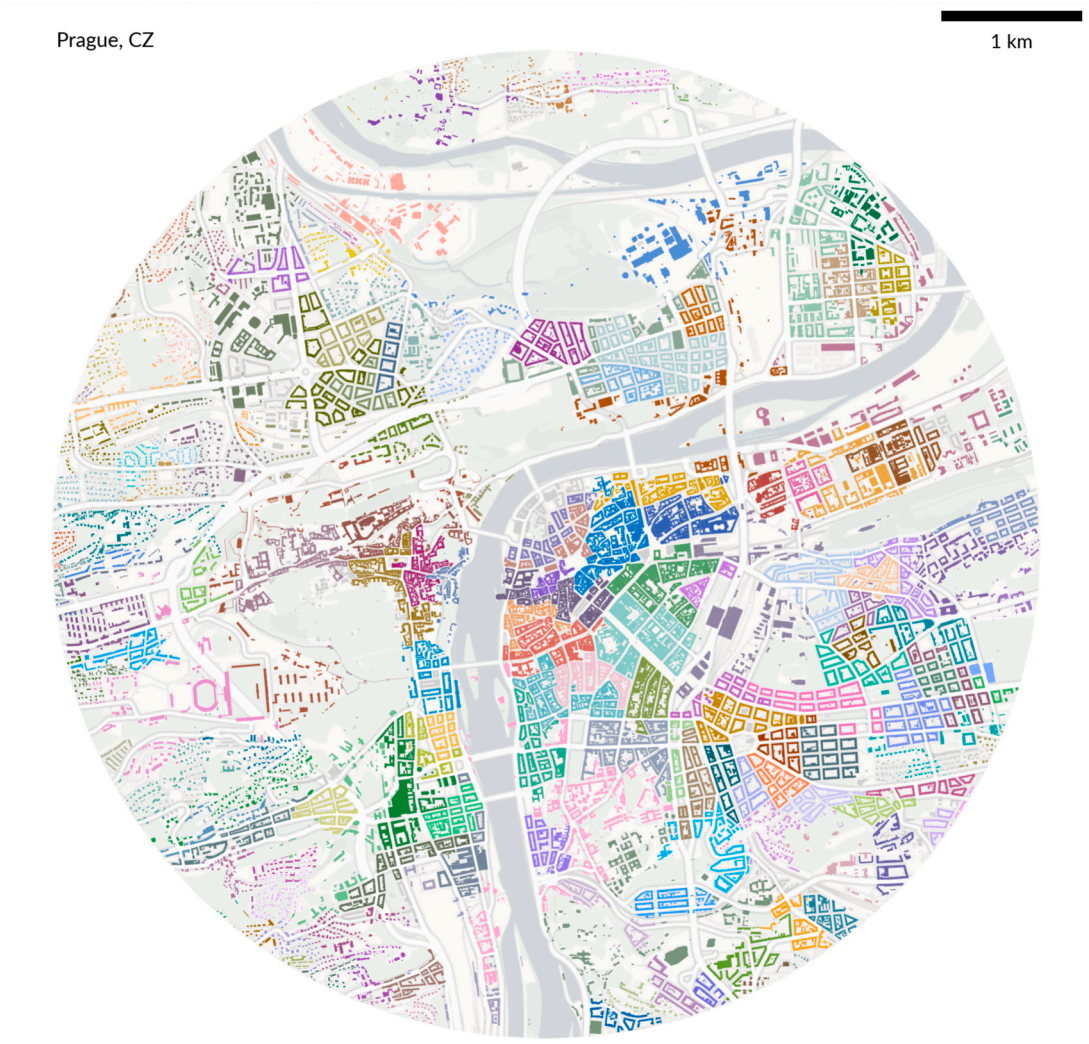


Fig. 2. Delineation of morphotopes in the city centre of Prague. Each color captures a single morphotope, while a light grey reflects buildings marked as 'noise' by SA[3].

thus examples of what morphotopes within that branch tend to look like. Note that the morphotopes are not to scale due to their varying spatial extents.

5.3. Tree interpretation

We derive names and short pen portraits of each of the eight taxonomic branches until the third level of bifurcation. We do this to improve the subsequent analysis, interpretation and mapping of the different types of urban fabric from the branches. The naming is derived from an expert interpretation of the morphometric profile of individual branches and attempts to stick strictly to this input. While it would be tempting to call certain branches *"medieval city"* or *"industrial area"*, neither historical origin nor land use played any role in the classification and the fact that they are showing in the taxonomy is only a result of their distinct morphological patterns. See *Appendix 14* for the detailed branch profiles, short names and branch descriptions.

The first bifurcation is characterised by the split between what we call *Coherent* and *Incoherent* fabric. This split mirrors the one identified by Dibble et al. (2019) and largely captures the distinction between places where all buildings, streets, plots and blocks have their traditional structural role (e.g. perimeter block of 1800s, see Levy (1999)), and those where at least one of is not significant from a structural perspective (e.g. modernist housing). It reflects the differences between pre-(coherent) and post-war (incoherent) development as well as those between traditional residential areas (coherent) and industrial zones (incoherent). Village development, where a block is only loosely defined, falls typically within the *Incoherent* branch, whereas garden cities fall typically within the *Coherent* branch.

In the second bifurcation, the *Coherent* fabric splits into *Dense* and *Interconnected*, reflecting various degrees of amalgamation, while the *Incoherent* fabric into *Large-scale* and *Small-scale*, capturing the varying scale of constituent elements. The third bifurcation then distinguishes between *Coherent Dense Disjoint* and *Coherent Dense Adjacent* fabrics based on the changes in the degree of building adjacency, between *Incoherent Large-scale Homogeneous* and *Incoherent Large-scale Heterogeneous*, which is driven by internal diversity of buildings and their configurations, and between *Linear*, *Sparse*, and *Compact* branches of *Incoherent Small-scale* fabric, distinguishing different types of typically rural and peri-urban development.

Fig. 4 shows how these splits manifest in four settlements of different sizes. This example shows that despite significant differences in the nature and geographical location of these places, similar morphotopes are consistently linked together and remain within the same branches of taxonomy.

We can further explore the third bifurcation, with eight named branches in Fig. 5, illustrating the outcomes of the classification across major cities in the region. City centres of all major cities fall under

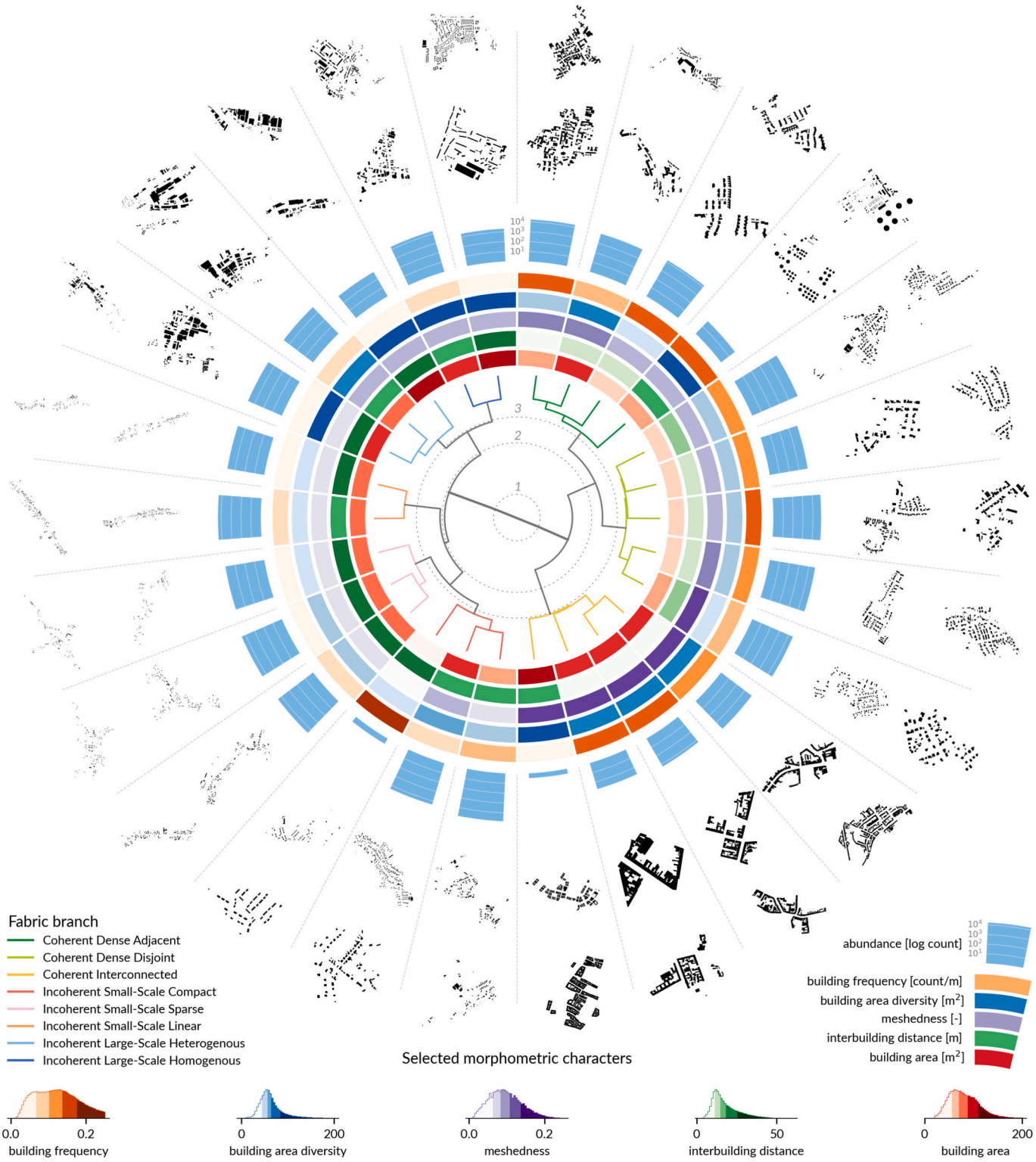


Fig. 3. Overview of the resulting hierarchical morphotope taxonomy, showing the radial representation of first levels of bifurcation the taxonomic tree (indicated by dashed lines), a sample of their morphometric profile, abundance and two example morphotopes for each branch.

Coherent Interconnected Fabric, with different extents, from the very small historical core of Bratislava to the large portion of Vienna. The composition of these cities vary due to many reasons. For example, both Prague and Bratislava show high proportions of *Incoherent Large-scale Homogeneous Fabric*, where we typically see large housing developments of the communist period. While some areas falling under the same fabric are present in Munich, and marginally in Vienna, they are much scarcer. Conversely, Vienna, and especially Munich, showcase a much larger proportion of *Coherent Dense Disjoint Fabric*, often covering developments in garden city styles or urban villas.

However, the taxonomy does not stop at the third level of bifurcation but can go arbitrarily granular. To illustrate that, Fig. 6 shows more granular, unnamed taxonomy branches for the city of Prague, CZ. The first column shows that Prague at the third level of bifurcation, similarly

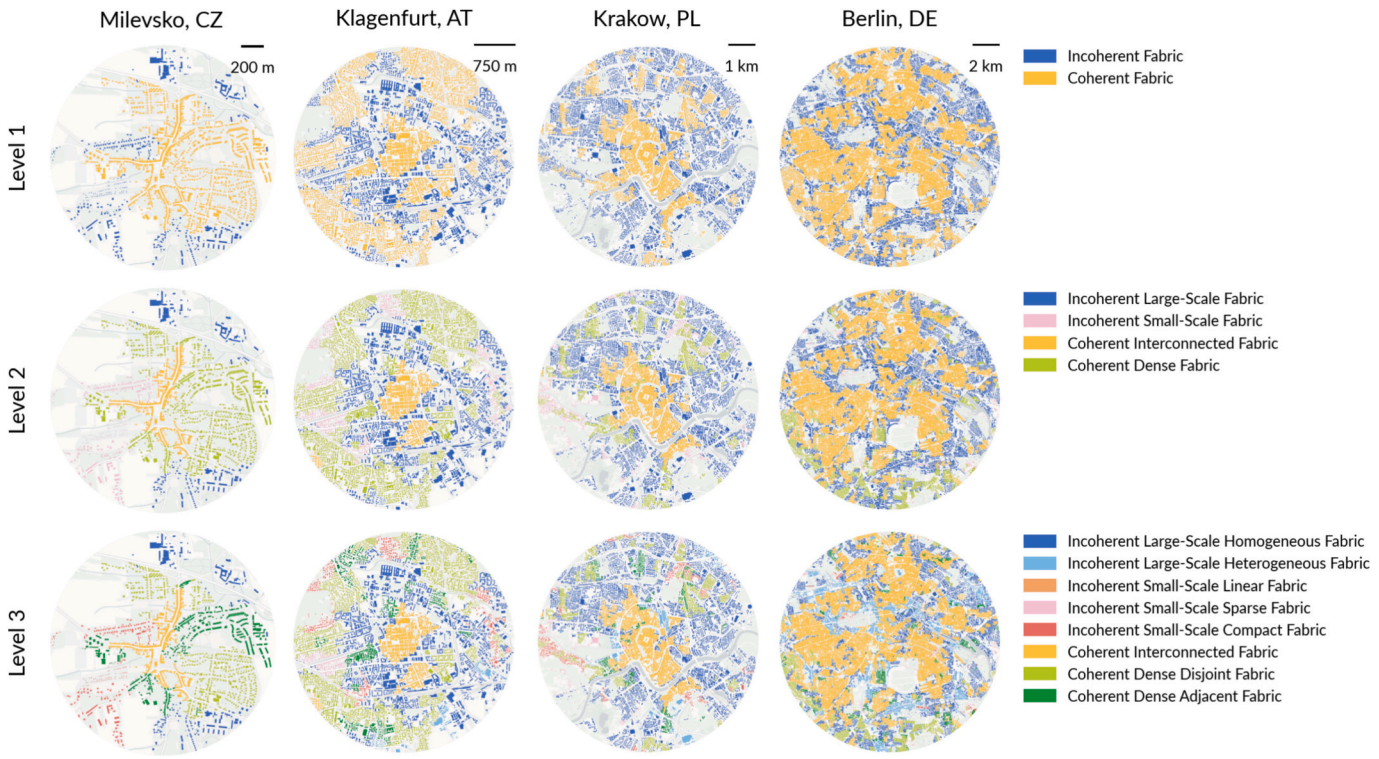


Fig. 4. Examples of delineation of urban fabric types at different levels of taxonomic granularity and across places of different scale.

to the cities and towns present in Fig. 4. The next two columns represent more granular cuts. In the second column, the *Coherent Interconnected Fabric* is split into two clusters, one of which has a more organic street pattern and buildings which form larger blocks, while the other has a more regular grid pattern and the buildings form smaller blocks with fewer internal courtyards. The former cluster is further split in column three – smaller blocks with organic street patterns in one cluster, and larger buildings in interconnected blocks in the other. Similarly, the latter cluster with the grid pattern, is split based on street characteristics and courtyard presence.

Each bifurcation is driven by a different subset of morphometric attributes. Their relative importance for the first 1000 bifurcations is reported in *Section 20*.

5.4. Spatial distribution of the results

Having a consistently derived, flexible taxonomy of this sort, covering multiple countries, enables multiple new types of analysis. While it is not our ambition to cover those within this publication, we analyse the spatial distribution of individual branches to showcase the scale, flexibility and potential applications of the taxonomy.

Table 1 shows the statistical distribution of the named clusters per country, across the first three levels. At level 1, we can observe a disbalance and radical difference between countries. For example, Poland and Lithuania are composed of *Incoherent Fabric* from more than 80%, while Germany only 30%. However, a subdivision to level 2 shows that this is due to high proportion of *Incoherent Small-Scale Fabric*, meaning typically villages and suburbia. Another detail can be uncovered at level 3. *Incoherent Large-Scale Homogeneous Fabric*, *Incoherent Large-scale Heterogeneous Fabric*, *Coherent Interconnected Fabric* are present in all countries in similar proportions, while *Incoherent Small-scale Linear Fabric* is more prominent in Poland and Lithuania. *Incoherent Small-scale Sparse Fabric* is present in all countries at around 30%, apart from Germany and Slovakia. Inversely, around half the urban form in Germany and Slovakia is *Coherent Dense Disjoint Fabric*, while it is at around 15% in the other countries. *Incoherent Small-Scale Compact Fabric* is present everywhere at different proportions, while *Coherent Dense Adjacent Fabric* is more prominent in Czechia, Austria and Germany.

Fig. 7 shows these proportions on sub-national scale, using a rectangular grid of 50 km edge size. What is shown further reiterates what is present in the Table 1 but indicates that the abundance of individual types of built-up fabric varies also within countries, capturing their internal heterogeneity.

5.5. Comparisons

A quantitative comparison between HiMoC and existing data products can highlight the similarities and the uniqueness of the proposed classification method. We offer such a comparison in the *Appendix 16*, based on co-occurrence, with three data products: CORINE Land Cover, Copernicus Urban Atlas and Local Climate Zones. In general, the branches of built-up fabric broadly follow the higher level patterns that are described by all three of data products. However, our branches tend to split their classes into different sub-classes, providing more information on internal structure of cities. One clear outcome of this comparison is that it seems that the large differences are due to different concepts being taken into account when making the classification. Given HiMoC is deeply embedded in morphological theory, it raises questions about the validity of these classifications, especially Local Climate Zones that are being used for this purpose, as a proxy for urban morphology outside of its original purpose (e.g. climate studies), as is being currently often used (Debray et al., 2025; Taubenböck et al., 2020).

6. Discussion & conclusion

The *Hierarchical Morphotope Classification* represents another step in evolution of morphometric classification methods. One, that builds upon previous research (Araldi & Fusco, 2024; Arribas-Bel & Fleischmann, 2022; Fleischmann et al., 2022) and pushes its limits to allow capturing a much wider geographical extent (theoretically, we are limited only by data availability and consistency), while providing higher precision, significantly improved flexibility and a direct connection to

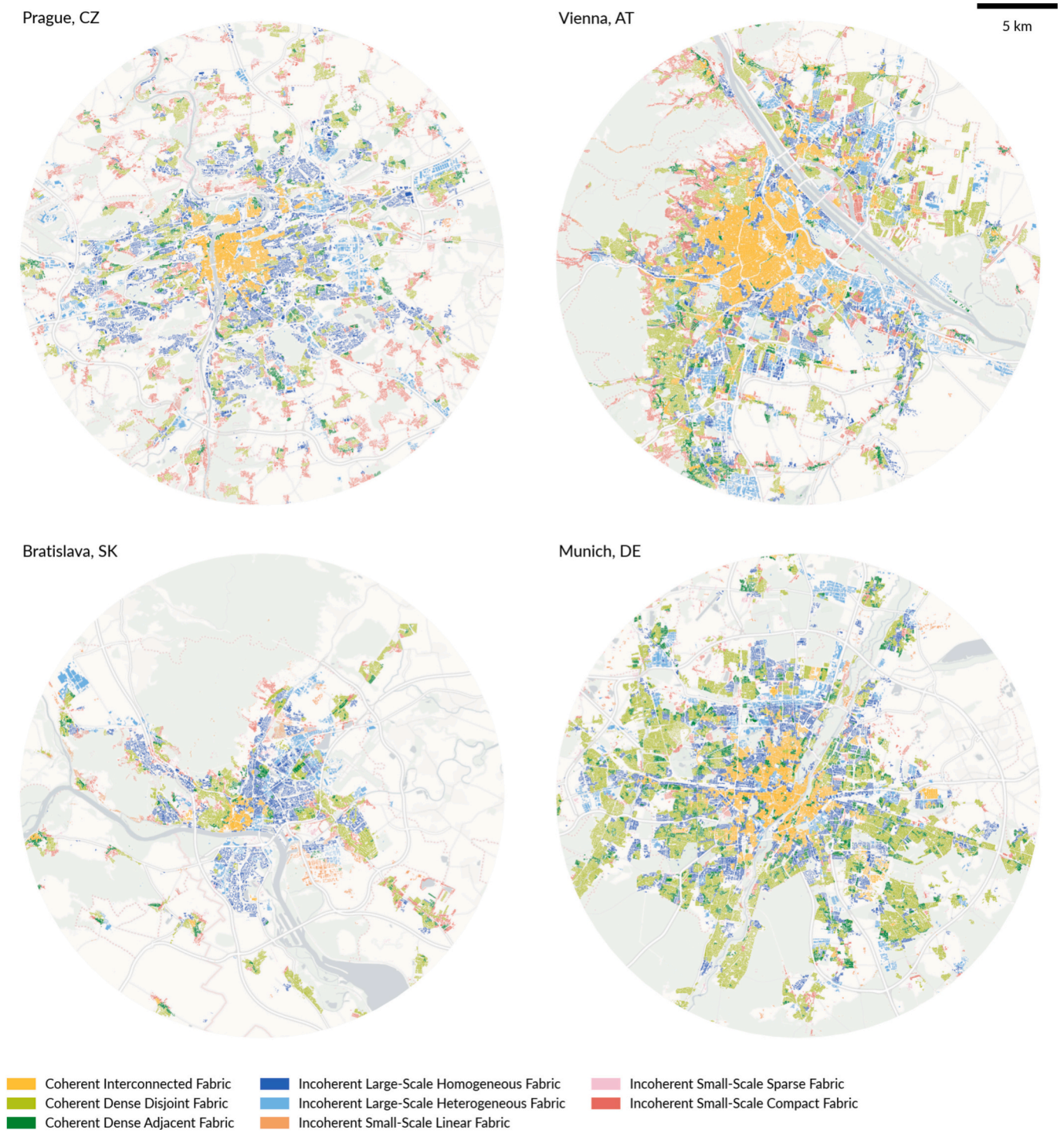


Fig. 5. Taxonomy cut at level 3 illustrated in Prague, CZ, Vienna, AT, Bratislava, SK, and Munich, DE.

morphological theory.

Previous research has been limited to individual cities and metropolitan areas (e.g. Araldi & Fusco, 2019; Fleischmann et al., 2022), with occasional attempts to capture the fabric of an entire nation (Araldi & Fusco, 2024; Fleischmann & Arribas-Bel, 2022). HiMoC moves the limit of scalability (see notes on scalability in *Section 17*) beyond that, allowing consistent analysis of built-up fabric across multiple countries.

Its hierarchical structure allows a high degree of flexibility when using the classification within custom use cases. The resolution that is suitable for a high-level overview of large metropolitan areas might use 8 classes, while one that is required for detailed analysis of a single city might need to work with 80, or an advanced ecological analysis of presence of types of built-up fabric might rely on hundreds. All of which are derived in a consistent quantitative manner. Given that the three coarsest levels come with a description and a pen-portrait, and every branch has a rich numeric profile, the flexible classification retains a high degree of interpretability at the same time.

The reliance on the morphotope concept and its operationalisation through the SA^3 regionalisation method comes with multiple benefits. First is an embedding of the core concept in Conzen's theory of urban

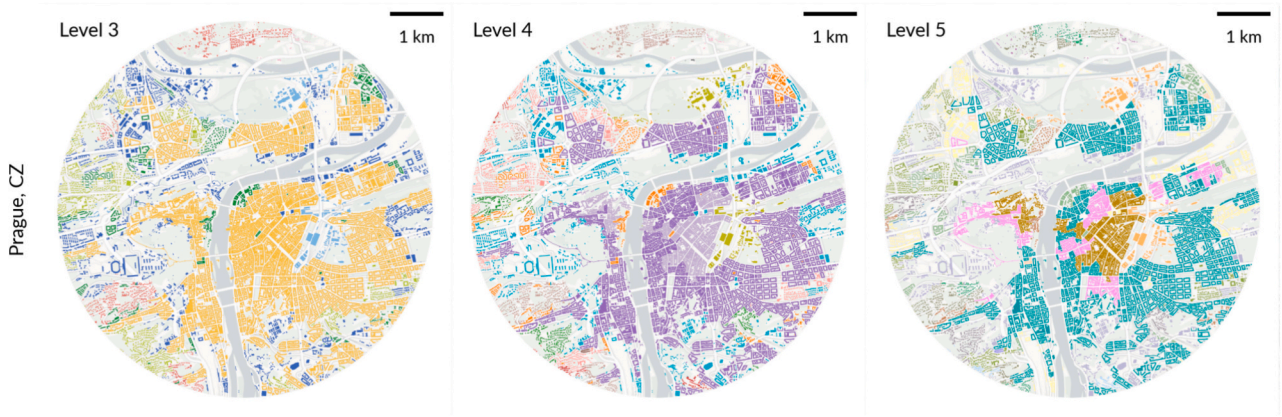


Fig. 6. Deeper cuts through the taxonomy within Prague, CZ show that we can get arbitrarily granular classification all the way down to the level of individual morphotopes.

Table 1
Distribution of fabrics at levels 1–3 per country in thousands of buildings. The value in brackets shows the proportion of buildings of a specific type to total buildings of a country.

	AT	CZ	DE	LT	PL	SK
(a) Level 1						
Incoherent Fabric	3153 (0.72)	2758 (0.67)	15,623 (0.30)	1671 (0.81)	12,860 (0.84)	1550 (0.45)
Coherent Fabric	1209 (0.28)	1372 (0.33)	35,807 (0.70)	381 (0.19)	2509 (0.16)	1885 (0.55)
(b) Level 2						
Incoherent Large-Scale Fabric	229 (0.05)	351 (0.09)	3722 (0.07)	90 (0.04)	1069 (0.07)	200 (0.06)
Incoherent Small-Scale Fabric	2923 (0.67)	2407 (0.58)	11,900 (0.23)	1580 (0.77)	11,790 (0.77)	1349 (0.39)
Coherent Interconnected Fabric	113 (0.03)	81 (0.02)	1129 (0.02)	2 (0.00)	76 (0.01)	25 (0.01)
Coherent Dense Fabric	1096 (0.25)	1291 (0.31)	34,677 (0.67)	378 (0.18)	2432 (0.16)	1859 (0.54)
(c) Level 3						
Incoherent Large-scale Homogeneous Fabric	132 (0.03)	247 (0.06)	1911 (0.04)	67 (0.03)	780 (0.05)	137 (0.04)
Incoherent Large-scale Heterogeneous Fabric	97 (0.02)	104 (0.03)	1811 (0.04)	23 (0.01)	289 (0.02)	63 (0.02)
Incoherent Small-scale Linear Fabric	223 (0.05)	226 (0.05)	1937 (0.04)	452 (0.22)	4156 (0.27)	385 (0.11)
Incoherent Small-scale Sparse Fabric	1500 (0.34)	1102 (0.27)	3149 (0.06)	621 (0.30)	5122 (0.33)	234 (0.07)
Incoherent Small-scale Compact Fabric	1199 (0.27)	1078 (0.26)	6813 (0.13)	506 (0.25)	2510 (0.16)	730 (0.21)
Coherent Interconnected Fabric	113 (0.03)	81 (0.02)	1129 (0.02)	2 (0.00)	76 (0.01)	25 (0.01)
Coherent Dense Disjoint Fabric	788 (0.18)	681 (0.17)	28,918 (0.56)	369 (0.18)	2024 (0.13)	1768 (0.51)
Coherent Dense Adjacent Fabric	308 (0.07)	609 (0.15)	5759 (0.11)	8 (0.00)	408 (0.03)	91 (0.03)

morphology, ensuring that the method does not simply rely on some data output but that the output links to proved and verified ideas. Second is the fact that the regionalisation produces strictly contiguous boundaries, eliminating a degree of fuzziness present in methods with fewer spatial restrictions (Arribas-Bel & Fleischmann, 2022; Fleischmann et al., 2022). Last is the adaptation to varying degrees of internal heterogeneity of built fabric. A method that treats heterogeneity uniformly typically gives too much space to locations where it is high, squashing the richness of relatively more homogeneous locations within the taxonomy.

While the taxonomic tree on top of morphotopes is conceptually similar to Conzen's hierarchy of morphological regions, we purposefully do not claim that it reflects it. The reason is a methodological difference between how morphological regions and their hierarchy are typically derived and how we derive our hierarchy. While this difference is still present when dealing with morphotopes, on such a small scale it is typically not visible. When going up the hierarchy, the difference between morphometric assessment of similarity and that of historical understanding of place grows. The two approaches start from the same idea but conceptualise it differently.

What the analysis of the taxonomy, as well as the morphometric profiling in *Section 14*, shows is another key aspect of the work – the fact that the results were based on both building and street characteristics. If attributes derived from one or the other were concentrated at the top or at the bottom of the hierarchy, it would suggest that only one set dominates the dissimilarity between the fabrics. However, that is not the case and both building and street patterns tend to affect the structure of the taxonomy at multiple levels. In fact, the first bifurcation is between, on the one hand, built fabric that closely follows the traditional relationship between buildings, plots and streets, and on the other, built fabric which breaks that relationship. Based on this result, classifications that focus only on streets or buildings in isolation are missing morphological information and potentially produce delineations at coarser scales of dissimilarity than is offered here.

Yet, the method comes with a set of limitations. At the moment, its applicability is restricted to areas where we have (relatively) consistent data inputs. Even though we rely only on 2-dimensional building footprint polygons and street networks, the building data is not always available and when it is, it might be inconsistent with INSPIRE (e.g. Microsoft Building Footprints) or even heterogeneous (e.g. OpenStreetMap and its derivatives like Overture Maps). Combining such data is not really possible as the morphometric attributes derived from different representations of buildings is not comparable. In the same vein, while the reliance on 2D data increases applicability, the method does not take into account height which might help distinguish certain branches, especially those containing modernist housing and utilitarian developments, where both showcase similar 2D patterns but are very different height-wise.

Furthermore, the stability of the taxonomic tree presented here is an open question. When extending the classification to cover additional countries, the tree will inevitably reshuffle, especially when moving culturally further away from current study area. Having stable branches at least in the coarsest levels would be desirable and strengthen the

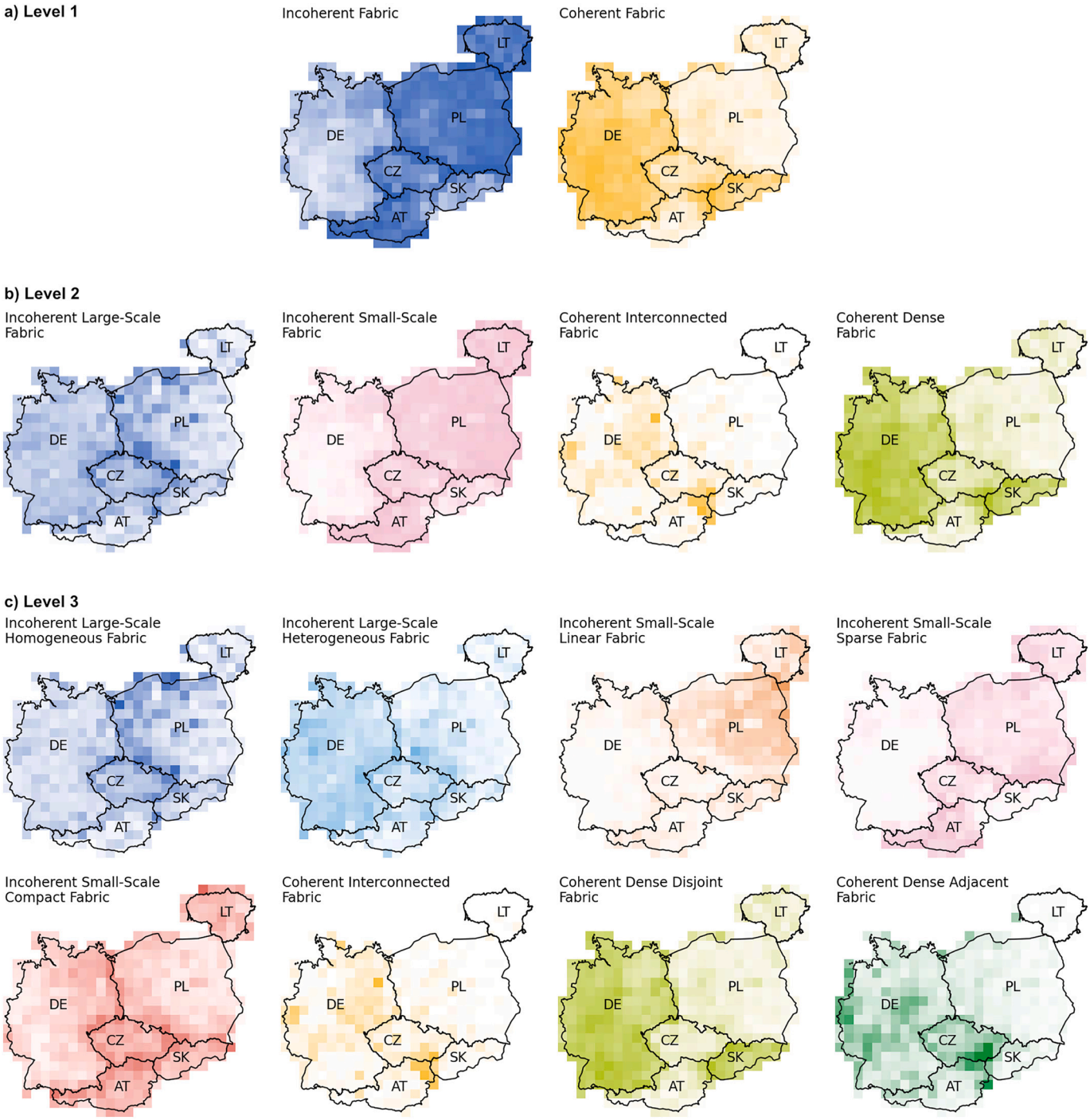


Fig. 7. Relative abundance of branches at level 1 indicating presence of different fabrics in different geographical regions. The opacity of each color reflects the relative presence in each cell, compared to the observed maximum, showcasing spatial distribution of individual branches.

results of this work, but whether that is the case, we do not know at the moment.

To address this question, future work shall focus on expansion of the coverage of the classification, both in Europe and outside. Many additional European countries offer data falling under the INSPIRE standard (e.g. France, Spain, The Netherlands, Belgium) and it is expected that these will be included in the classification in the upcoming years. Efforts like EUBUCCO (Milojevic-Dupont et al., 2023) might help with this progress by eliminating the need to fetch the data from individual countries but there still needs to be a robust framework for assessment of data consistency. Similar data are occasionally available outside Europe as well, as in case of Medellin or Singapore, for example. Where the data is not available in the expected form, another branch of future work shall focus on gap filling methods using other types of data, be it satellite imagery or sub-optimal data coming from Overture Maps or other sources. At the same time, this classification reflects one point in time, without a temporal dimension. This is yet again mostly due to data availability, but we may look into a method based on satellite imagery that inherently comes as a time series to fill this dimension (Fleischmann & Arribas-Bel, 2024).

Another fruitful research avenue shall focus on extraction of insights from the classification itself, relating it to the effect of political

boundaries on the structure of settlements, environmental properties of different fabrics or relation of population groups and fabric types.

We need to better understand how our settlements are structured, what has formed them and how they affect the human population, environment or economy. None of this can be done without methods enabling analysis that are theory-driven, detailed, consistent and scalable. Given the complexity of urban morphology, we can expect that classification approaches, reducing the complexity to a small number of manageable and interpretable classes, will not fade away. HiMoC offers one such classification method, that aims to look at one specific slice of human settlements, their built form, and unpack it into a flexible hierarchical tree built on top of morphotopes, the smallest locality obtaining distinctive character among their neighbours (Conzen, 1988). We believe that its complementarity with other existing large-scale classification methods paves the way for ubiquitous applicability beyond urban morphology and morphological analysis.

CRediT authorship contribution statement

Martin Fleischmann: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Krasen Samardzhiev:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Anna Brázdová:** Writing – review & editing, Validation. **Daniela Dančejová:** Writing – review & editing, Validation. **Lisa Winkler:** Writing – review & editing, Validation, Data curation.

Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cities.2026.107047.

Data availability

Data is publicly available and linked from the paper (title page).

All components of the work rely on open source software and open data, with the resulting code and data being openly available. Code, with the specification of a reproducible environment, is available at https://github.com/uscuni/urban_taxonomy and archived at https://doi.org/10.5281/zenodo.17105270. The classification is released as an open data product available at https://urbantaxonomy.org and https://doi.org/10.5281/zenodo.17076283. The SA^3 functionality has been contributed to the open source package spopt (Feng et al., 2022), morphometric assessment to momepy (Fleischmann, 2019), and features required to support those to libpysal (Rey et al., 2022), shapely (Gillies et al., 2025) and geopandas (Van den Bossche et al., 2025).

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